

PHY5505: DATA ACQUISITION AND PROCESSING SKILLS FOR PHYSICISTS II

Effective Term

Semester A 2026/27

Part I Course Overview

Course Title

Data Acquisition and Processing Skills for Physicists II

Subject Code

PHY - Physics

Course Number

5505

Academic Unit

Physics (PHY)

College/School

College of Science (SI)

Course Duration

One Semester

Credit Units

3

Level

P5, P6 - Postgraduate Degree

Medium of Instruction

English

Medium of Assessment

English

Prerequisites

Nil

Precursors

Nil

Equivalent Courses

Nil

Exclusive Courses

Nil

Part II Course Details

Abstract

This course is second part of Data Acquisition and Processing Skills for Physicists I and II. The primary aim of this course is to equip physics students with fundamental skills in data acquisition and processing, thereby enhancing their ability to confidently handle diverse data sources. These skills are essential for modern physicists who must be adept not only in theoretical calculations but also in the practical aspects of data handling. Students will learn basic data analysis methods and utilize various programming languages, with a focus on Python, to process and visualize data. Additionally, the course will cover the use of various hardware tools for data collection, including but not limited to microcontrollers like Arduino, sensors, and real-time data processing units. The students are expected to finish several mini-projects on data acquisition and processing. This course is implemented based on student projects.

Course Intended Learning Outcomes (CILOs)

CILOs		Weighting (if app.)	DEC-A1	DEC-A2	DEC-A3
1	Be able to acquire and understand data from different sources	50	x	x	
2	Be able to process big data sets	50	x	x	

A1: Attitude

Develop an attitude of discovery/innovation/creativity, as demonstrated by students possessing a strong sense of curiosity, asking questions actively, challenging assumptions or engaging in inquiry together with teachers.

A2: Ability

Develop the ability/skill needed to discover/innovate/create, as demonstrated by students possessing critical thinking skills to assess ideas, acquiring research skills, synthesizing knowledge across disciplines or applying academic knowledge to real-life problems.

A3: Accomplishments

Demonstrate accomplishment of discovery/innovation/creativity through producing /constructing creative works/new artefacts, effective solutions to real-life problems or new processes.

Learning and Teaching Activities (LTAs)

LTAs		Brief Description	CILO No.	Hours/week (if applicable)
1	Projects	The students will complete several mini-projects related to data acquisition and processing.	1, 2	
2	Presentations/exhibition	The students will showcase projects.	1, 2	3 Hours

Assessment Tasks / Activities (ATs)

ATs		CILO No.	Weighting (%)	Remarks ("- for nil entry)	Allow Use of GenAI?
1	Presentations/exhibition	1, 2	100	-	Yes

Continuous Assessment (%)

100

Assessment Rubrics (AR)

Assessment Task

Presentations/exhibition

Criterion

Demonstrating the capability in data acquisition and processing.

Excellent (A+, A, A-)

High (excellent accomplishment with creativity and correct understanding)

Good (B+, B, B-)

Significant (good accomplishment with mostly correct understanding)

Fair (C+, C, C-)

Moderate (fair accomplishment with some correct understanding)

Marginal (D)

Basic (essential accomplishment with basic understanding)

Failure (F)

Not reaching marginal level

Part III Other Information

Keyword Syllabus

- Introduction to data
- Data acquisition
- Data structure, database, data acquisition hardwires, e.g., Arduino.
- Data analysis methods, e.g., linear regression, bootstrap and jackknife
- Data processing, e.g., Markov chain Monte Carlo
- Data processing program languages, e.g., Python, Matlab, Jupyter, Data Visualization, e.g., Matplotlib.

Reading List**Compulsory Readings**

Title	
1	None

Additional Readings

Title	
1	Z. Ivezic, A. Connolly, J. T. VanderPlas, and A. Gray, Statistics, Data Mining, and Machine Learning in Astronomy, Princeton University Press, 2014
2	D. W. Hogg, Data analysis recipes, arXiv:0807.4820, 1008.4686, 1205.4446, 1710.06068, 2005.14199